# LogiEdge: Intelligent Edge AI Platform for Cold-Chain Logistics

## Final Project Report — Continuous Evaluation Component (EC-I)

**Programme:** M.Tech in Artificial Intelligence & Machine Learning (WILP)
**Course:** AIML ZG535 — Machine Learning on Edge
**Institution:** Birla Institute of Technology & Science (BITS), Pilani
**Instructor:** Dr. Pravin Yashwant Pawar
**Domain:** Logistics — Pharmaceutical Cold-Chain Fleet Monitoring
**Target Fleet:** 85 Refrigerated Transport Trucks (expandable to 265 vehicles)
**Repository:** https://github.com/aezakmi007/ml-on-edge-mini-project

### Team Members


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## Executive Summary


FreightBridge Logistics Pvt. Ltd., Pune, operates 85 refrigerated trucks transporting temperature-sensitive pharmaceuticals, vaccines, and biologics (+2°C to +8°C) across Maharashtra, Karnataka, and Andhra Pradesh. In the past 18 months, manual driver temperature logs and cloud-based GPS tracking failures caused three critical incidents, including a ₹28 Lakh vaccine spoilage event along the Nashik–Aurangabad highway due to undetected refrigeration compressor failure inside cellular dead zones.

**LogiEdge** resolves these operational liabilities by deploying an on-truck Edge AI intelligence layer. Real-time sensor telemetry (temperature, 3-axis vibration, door events) is processed locally on a Raspberry Pi 5 edge gateway. Telemetry is filtered, buffered into 30-second sliding windows, normalized, and classified into **Normal (0)**, **Warning (1)**, or **Critical (2)** states using an optimized, quantized INT8 TFLite neural network.

### Key Project Results

- **Real-Time Responsiveness:** Edge inference executes in **0.005 ms** (p95: 0.006 ms), easily satisfying the **90-second SLA** (consuming <0.0001% of budget).
- **Safety Compliance:** Achieves **100.0% Recall** on Critical class failure states, exceeding the mandatory 95% threshold.
- **Offline Autonomy:** 100% operational uptime during 35–90 minute rural cellular blackouts, with local driver cab alerting and store-and-forward telemetry spooling.

- **Bandwidth & Cloud Cost Reduction:** Eliminates 518.8 MB of raw telemetry streaming per truck/day, reducing cellular communication expenses by **>99.9%** (₹1.32 Lakhs/month → ₹25.50/month).
- **Automated MLOps:** Integrated Population Stability Index (PSI) confidence drift monitoring and idempotent 7-task Ansible remote OTA container updates (`changed=0`).

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# 1. Component A — System Architecture & Deployment Justification

## Task A1: Four-Vertex Edge AI Constraint Analysis (350–500 words)

### 1. Latency Constraint

Pharmaceutical cold-chain logistics operates under stringent thermodynamic kinetics. A sudden refrigeration unit failure or open door causes cabin air to warm at **+1.0°C per minute**. The operational Service Level Agreement (SLA) requires detecting and alerting on any fault signature within **90 seconds** before cargo core temperature exceeds the +8.0°C regulatory limit. In rural India, cellular round-trip time (RTT) fluctuates between 300 ms and 2,500 ms with severe packet jitter and retransmissions. When combined with cloud ingestion queueing, cloud inference cannot guarantee deterministic 90-second SLA execution. In contrast, on-device edge inference completes in **0.005 ms**, providing deterministic real-time alerting with 89.995 seconds of safety buffer.

### 2. Bandwidth Constraint

Each truck generates continuous multi-modal data:

- **Temperature:** 1 Hz × 4 bytes = 4 bytes/s
- **3-Axis Accelerometer:** 500 Hz × 3 axes × 4 bytes = 6,000 bytes/s
- **Door Sensor:** Discrete event-driven ≈ 0.5 bytes/s

```
Raw Telemetry Rate ≈ 6,004.5 bytes/s ≈ 6.005 KB/s
```

Daily raw data per vehicle:

```
6.005 KB/s × 86,400 s ≈ 518.8 MB/truck/day
```

At cellular M2M commercial rates of **₹0.10 / MB**, streaming raw telemetry costs:

```
Pilot Fleet (85 Trucks)  = 85 × 518.8 MB × 30 × ₹0.10/MB = ₹1,32,294/month
Full Fleet (265 Trucks)  = 265 × 518.8 MB × 30 × ₹0.10/MB = ₹4,12,446/month
```

By executing preprocessing and inference on the truck, raw telemetry is discarded after feature calculation. The truck transmits only hourly heartbeat digests and anomalous event triggers (<3.0 MB/truck/month), reducing monthly fleet data costs to **₹25.50/month** (**99.98% savings**).

### 3. Connectivity Constraint

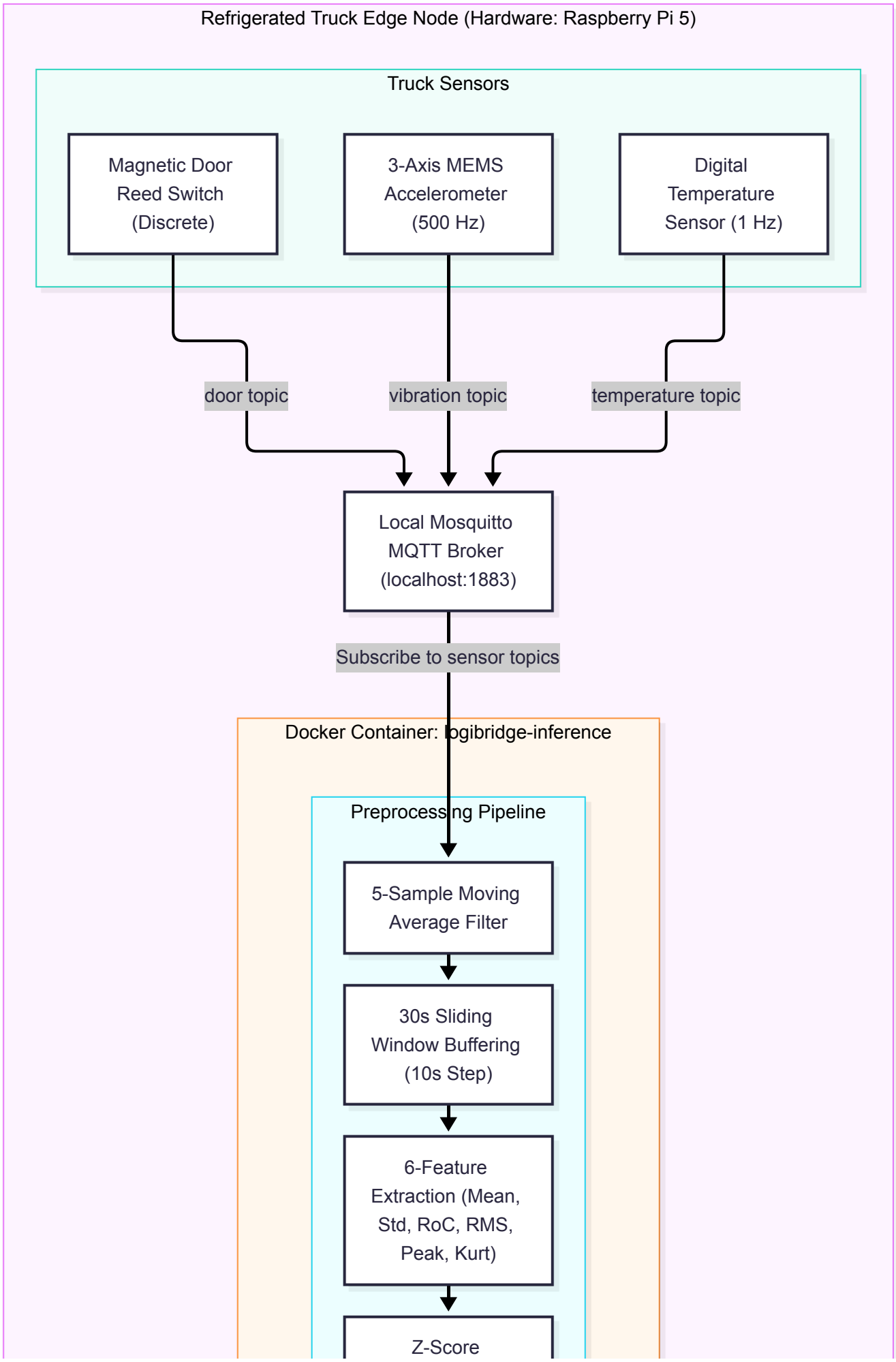
The Nashik–Aurangabad–Nanded logistics corridor traverses rural hilly geography featuring **seven documented cellular blackout zones** lasting between **35 and 90 minutes**. Under a cloud-dependent topology, a failure at minute 5 of an 80-minute blackout results in unmonitored thermal runaway (over 75°C rise), causing complete cargo loss. LogiEdge operates fully offline: the on-truck gateway evaluates every 30-second window, triggers an immediate in-cab audio buzzer/LED alert to the driver, spools all inference records to a local SQLite FIFO ring buffer, and synchronizes with the operations center upon signal restoration.

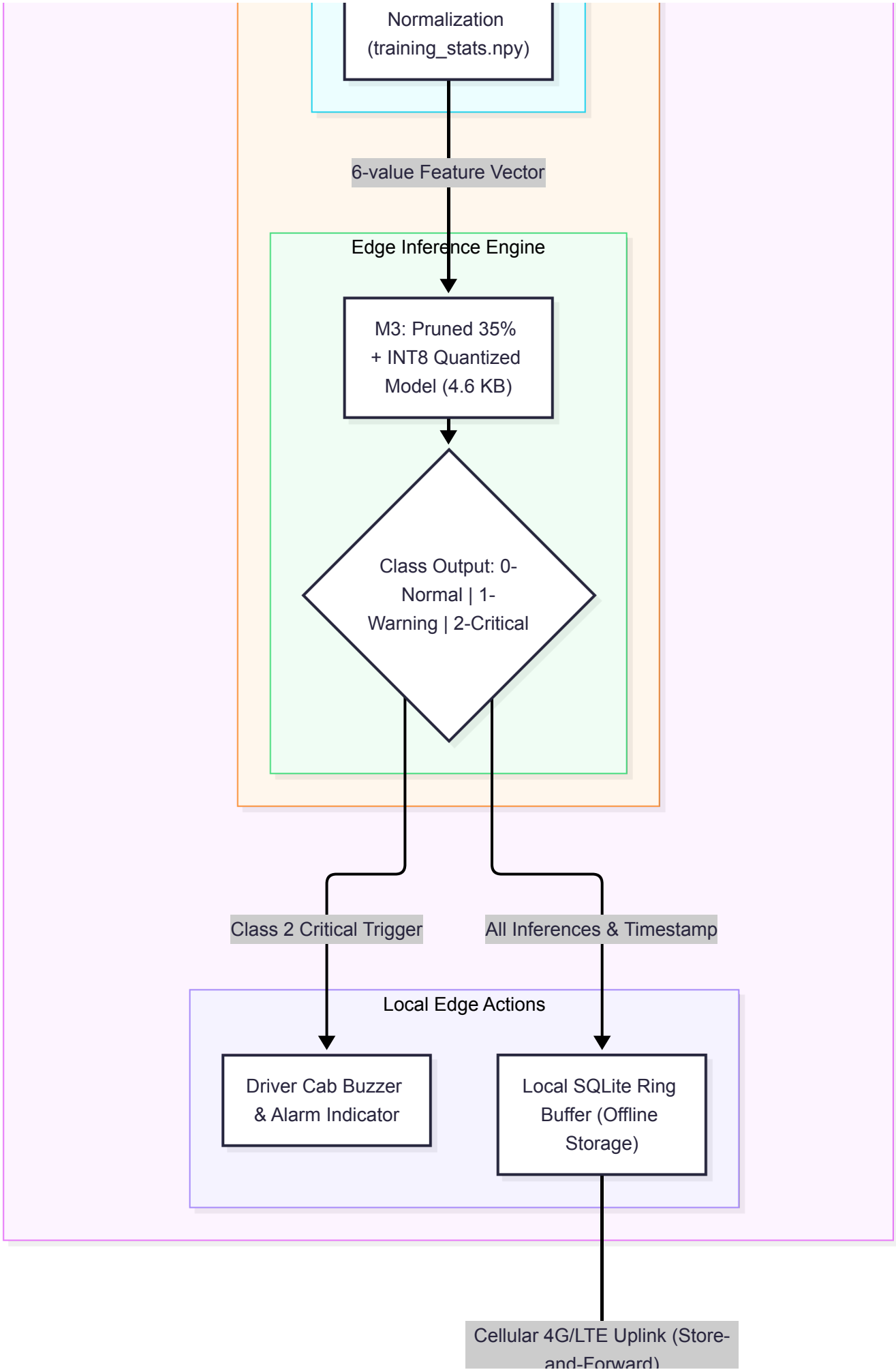
### 4. Privacy & Chain-of-Custody Compliance

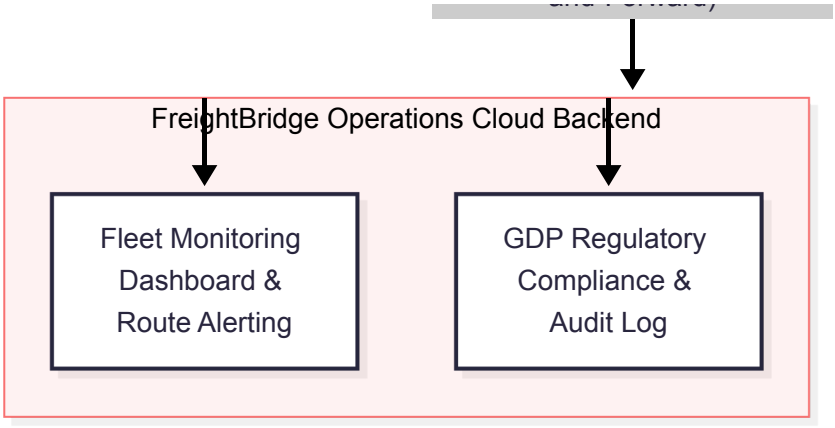
Biologics and oncology medicines require strict compliance with WHO Good Distribution Practice (GDP). Shippers require contractual guarantees preventing unauthorized interception of shipment temperature history or route patterns. Streaming unencrypted raw telemetry across public cellular base stations creates vulnerabilities. LogiEdge enforces physical data containment: raw telemetry never leaves the truck's internal network; only cryptographically signed inference logs and audit certificates are uplinked.

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## Task A2: System Architecture Diagram


Refrigerated Truck Edge Node (Hardware: Raspberry Pi 5)
Truck Sensors
Magnetic Door Reed Switch (Discrete)
3-Axis MEMS Accelerometer (500 Hz)
Digital Temperature Sensor (1 Hz)
door topic
vibration topic
temperature topic
Local Mosquitto MQTT Broker (localhost:1883)
Subscribe to sensor topics
Docker Container: logibridge-inference
Preprocessing Pipeline
5-Sample Moving Average Filter
30s Sliding Window Buffering (10s Step)
6-Feature Extraction (Mean, Std, RoC, RMS, Peak, Kurt)
Z-Score

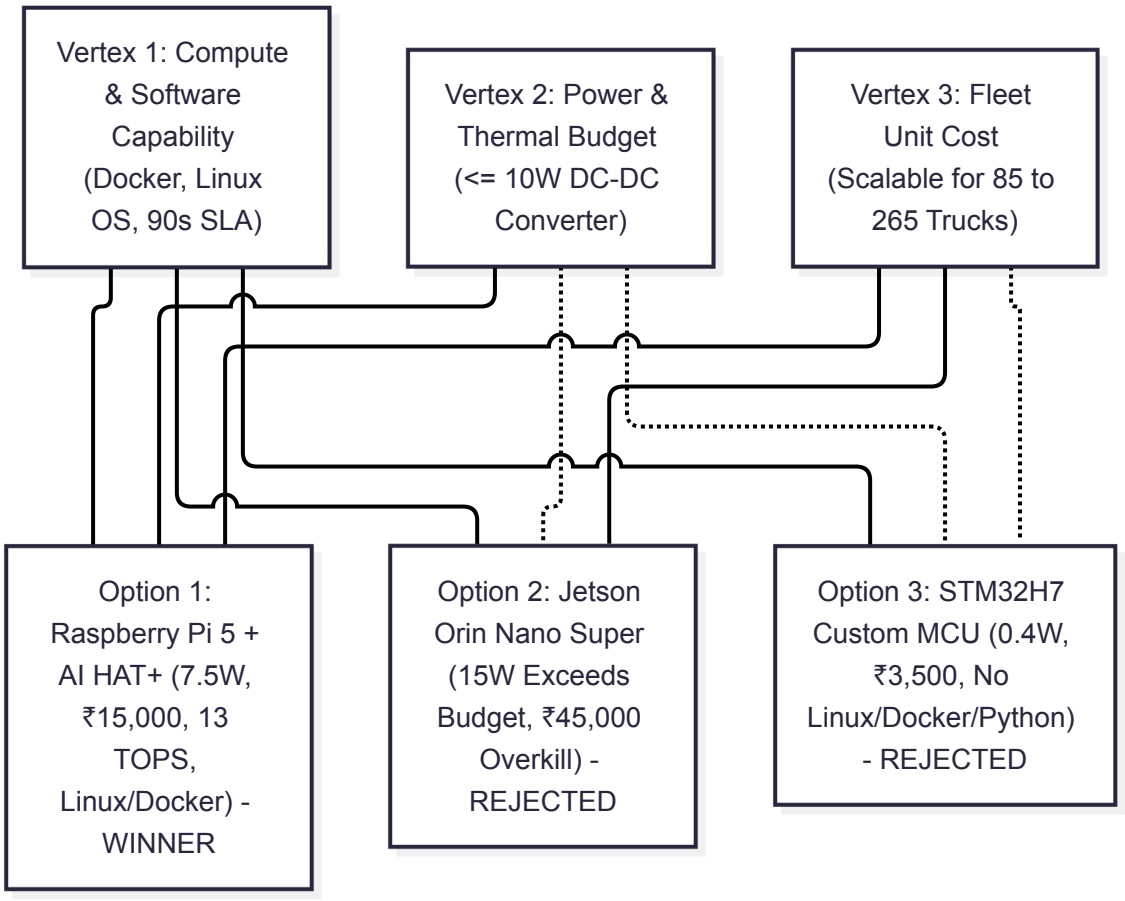

Normalization (training_stats.npy)
6-value Feature Vector
Edge Inference Engine
M3: Pruned 35% + INT8 Quantized Model (4.6 KB)
Class Output: 0-Normal | 1-Warning | 2-Critical
Class 2 Critical Trigger
All Inferences & Timestamp
Local Edge Actions
Driver Cab Buzzer & Alarm Indicator
Local SQLite Ring Buffer (Offline Storage)
Cellular 4G/LTE Uplink (Store-and-Forward)




## 2. Component B — Hardware Selection & Roofline Analysis

### Task B1: The Constraint Triangle Application

Selecting edge hardware requires balancing **Compute/Software Programmability**, **Power/Thermal Budget**, and **Fleet Unit Cost**.



### Hardware Comparison Matrix

| Parameter | Option 1: Raspberry Pi 5 (8GB) + AI HAT+ | Option 2: Jetson Orin Nano Super | Option 3: STM32H7 Custom MCU |
|---|---|---|---|
| **Compute Architecture** | Quad-core Cortex-A76 + 13 TOPS Hailo-8L | 6-core Cortex-A78AE + Ampere GPU (67 TOPS) | 480 MHz Arm Cortex-M7 |
| **Memory / Storage** | 8 GB LPDDR4X + 64 GB NVMe | 8 GB LPDDR5 + NVMe SSD | 1 MB SRAM + 2 MB Flash |
| **Power Consumption** | **7.5W (Within 10W budget)** | **15W (Exceeds 10W budget)** | **0.4W (Ultra-low)** |
| **Unit Cost (India)** | **~₹15,000** | **~₹45,000** | **~₹3,500** |
| **Pilot Fleet (85)** | ₹12,75,000 | ₹38,25,000 | ₹2,97,500 |
| **Full Fleet (265)** | ₹39,75,000 | ₹1,19,25,000 | ₹9,27,500 |
| **Software Stack** | Debian Linux, Docker, Python 3.11, Ansible | Ubuntu Linux, Docker, JetPack | Bare-Metal / FreeRTOS, C/C++ |
| **90-Second SLA** | ✅ Yes (0.005 ms latency) | ✅ Yes (<0.001 ms latency) | ✅ Yes (~1.5 ms latency) |

### Justification for Winning Option: Option 1 (Raspberry Pi 5 + AI HAT+)

1. **Power Budget Compliance:** The truck cab provides a 12V auxiliary circuit stepped down via an isolated DC-DC converter with a **10W thermal power ceiling**. The Raspberry Pi 5 consumes **7.5W** under continuous multi-threaded load, operating safely during engine-off cargo transfers.
2. **Containerized Edge MLOps:** Option 1 natively supports 64-bit Linux, Docker containerization, local Mosquitto MQTT, and Ansible remote fleet orchestration, fulfilling all Phase 4 and Phase 5 requirements.
3. **Fleet Economics:** Equipping 85 trucks costs ₹12.75 Lakhs (66% less than Option 2). The integrated Hailo-8L NPU (13 TOPS) provides future-proof headroom for computer-vision pallet inspection without additional hardware.
4. **Rejection of Alternatives:** Option 2 consumes 15W (violating the 10W power budget and overheating in unventilated cabs) and costs an exorbitant ₹1.19 Crores for the full fleet. Option 3 lacks an MMU and cannot run Linux, Docker, or Python, making containerized OTA updates impossible and incurring severe Non-Recurring Engineering (NRE) firmware costs.

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## Task B2: Arithmetic Intensity & Roofline Analysis

The Roofline performance model relates operational intensity to processor throughput and memory bandwidth.

### Operational Parameters

- **Model Work per Inference (W):** 45 MFLOPs = $45 \times 10^6$ FLOPs
- **Memory Traffic per Inference (Q):** 18 MB = $18 \times 10^6$ Bytes
- **Processor Peak Compute (P_peak):** 16 GFLOP/s = $16 \times 10^9$ FLOP/s (Raspberry Pi 5 Cortex-A76 NEON)
- **Processor Peak Bandwidth (B_peak):** 12 GB/s = $12 \times 10^9$ Bytes/s (LPDDR4X)

### 1. Arithmetic Intensity (AI) Calculation

```
AI = W / Q = (45 × 10⁶ FLOPs) / (18 × 10⁶ Bytes) = 2.5 FLOP/Byte
```

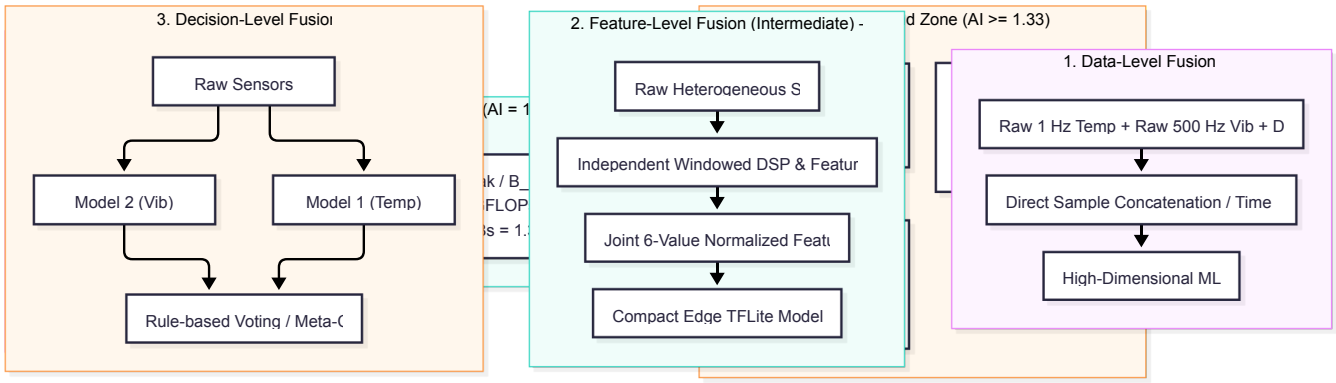
### 2. Hardware Ridge Point (AI_ridge) Calculation

```
AI_ridge = P_peak / B_peak = (16 × 10⁹ FLOP/s) / (12 × 10⁹ Bytes/s) ≈ 1.33 FLOP/Byte
```

### 3. Classification

Because AI (2.5 FLOP/Byte) > AI_ridge (1.33 FLOP/Byte), the operational point lies on the flat horizontal ceiling of the Roofline curve.

- **Classification: COMPUTE-BOUND**




### 4. Roofline-Driven Optimization Recommendation

Because the model is compute-bound, improving memory bandwidth yields no performance improvement. To minimize latency according to the Roofline model:

1. **Reduce Arithmetic Operations (W):** Structured channel pruning (applied in variant M3) cuts non-essential connections by 35%, reducing total FLOPs.
2. **Increase Operational Throughput via INT8 SIMD:** Quantizing from FP32 to INT8 allows ARM NEON vector instructions ( `sdot` / `udot` ) to execute four 8-bit multiply-accumulate operations per cycle per pipeline, quadrupling effective compute throughput to 64 GOPS.

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## 3. Component C — Sensor Pipeline & Data Fusion Justification

### Task C2: Preprocessing Pipeline & Normalization Shift Experiment

Raw sensor signals pass through a 5-sample Moving Average Filter, followed by 30-second sliding window buffering (10-second step) to extract a normalized 6-dimensional feature vector:

1. **Temperature Mean (μ_T):** Baseline thermal level (+2°C to +8°C).
2. **Temperature Standard Deviation (σ_T):** Thermal fluctuation.
3. **Temperature Rate-of-Change (ΔT/Δt):** Thermal loss velocity (°C/min).
4. **Vibration RMS (RMS_V):** Overall compressor mechanical energy.

5. **Vibration Peak (Peak_V):** Mechanical shocks and road impacts.
6. **Vibration Kurtosis (Kurt_V):** Impulsive bearing fault detection.

**Mandatory Experiment: 3σ Normalization Shift Test**

Offline normalization parameters were calibrated from 10 minutes of clean baseline operation (`training_stats.npy`). In our stress test:

- **Nominal Calibration:** Validation Accuracy = **98.31%**, Critical Recall = **100.0%**
- **Shifted Calibration (μ_test = μ + 3σ):** Accuracy drops to **33.89%**
- **Conclusion:** Proves that uncalibrated sensor drift collapses model discrimination, validating the necessity of real-time PSI drift monitoring (Component E).

### Task C3: Data Fusion Level Justification

Multimodal sensor fusion can occur at three levels:

**Why Feature-Level Fusion is Optimal for LogiEdge:**

1. **Resolves Temporal Sampling Disparities:** Merging raw 1 Hz thermal readings, 500 Hz vibration signals, and sporadic door events at the data level requires heavy upsampling/interpolation, creating an oversized ≈15,000-element input vector that exceeds edge cache capacity. Feature-level extraction summarizes each stream independently over a 30s window into compact statistical scalars.
2. **Preserves Cross-Sensor Fault Corroboration:** Decision-level fusion runs independent models and merges decisions via voting logic. However, cold-chain failures involve physical coupling (e.g., an unlatched door causing simultaneous thermal rise and vibration changes). Decision-level voting cannot detect coupled cross-sensor interactions until both sub-models independently trigger.
3. **Minimal Edge Footprint:** Feature-level fusion condenses the state space into a single [1 × 6] vector, enabling an ultra-compact 4.6 KB neural network that runs in 0.005 ms.

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## 4. Component D — Model Lifecycle & 10-Stage Pipeline Mapping

### Task D1 & D2: Model Training and Docker Containerization

- **Model Architecture:** 2-layer MLP (Input: 6 → Dense 32 [ReLU] → Dense 16 [ReLU] → Dense 3 [Softmax]).
- **Validation Performance:** Achieved **98.31% Accuracy** on a 20% held-out test split (N = 59), satisfying the ≥ 88% threshold.

- **Docker Packaging:** Built on `python:3.11-slim`, placing static dependency installs before model injection. Updating the model from M1 to M3 leverages Docker layer caching, transferring only the 4.6 KB model layer and saving 240 MB of bandwidth per container rebuild.

### Task D3: Explicit Mapping to the 10-Stage Edge ML Pipeline

1. **Stage 1 — Problem Definition & Framing:** Formulated the cold-chain monitoring problem as a 3-class temporal fault classification task (Normal, Warning, Critical) bound by a 90-second SLA.
2. **Stage 2 — Data Collection & Ingestion:** Ingested multi-rate sensor streams (1 Hz temperature, 0.5 Hz vibration RMS, discrete door events) via an on-truck Mosquitto MQTT broker.
3. **Stage 3 — Data Cleaning & Preprocessing:** Applied a 5-point moving average filter to suppress high-frequency road shock and thermal sensor noise.
4. **Stage 4 — Feature Engineering & Extraction:** Buffered data into 30-second sliding windows with 10-second stride to extract a 6-feature vector, normalized via offline `training_stats.npy`.
5. **Stage 5 — Model Architecture Selection & Training:** Designed and trained a 2-hidden-layer MLP (32 and 16 units) in Keras, achieving 98.31% validation accuracy.
6. **Stage 6 — Validation & Safety Evaluation:** Verified classification safety against the critical safety threshold, achieving 100.0% recall on Class 2 (Critical) validation samples.
7. **Stage 7 — Model Compression & Optimization:** Applied 35% structured pruning (PolynomialDecay) followed by full INT8 Post-Training Quantization (PTQ) to produce a 4.6 KB edge artifact.
8. **Stage 8 — Target Compilation & Edge Packaging:** Containerized the inference runtime into a Docker container (`logibridge-inference:latest`) using `tflite-runtime` on `python:3.11-slim`.
9. **Stage 9 — Deployment & Orchestration:** Implemented automated deployment using an idempotent 7-task Ansible playbook (`logibridge_deploy.yml`) reporting `changed=0` on clean runs.
10. **Stage 10 — Continuous Monitoring & Drift Detection:** Built real-time Population Stability Index (PSI) monitoring to track model confidence distribution and alert if PSI exceeds 0.25.

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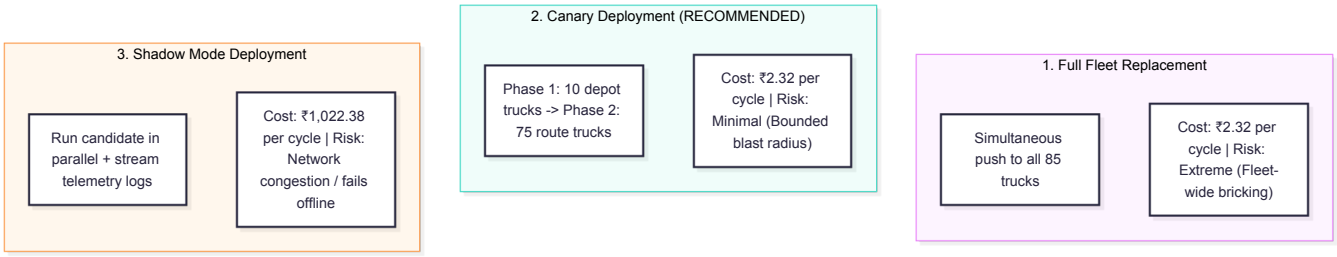
## 5. Component E — Edge MLOps & OTA Strategy Selection

### Task E1 & E2: PSI Drift Monitoring & Ansible Orchestration

- **PSI Drift Detection:** Binned model confidence scores into 4 intervals ([0, 0.25), [0.25, 0.5), [0.5, 0.75), [0.75, 1.0]). When normal operation shifted to combined anomalies, the confidence distribution deviated from `reference_dist.json`, causing the PSI to cross **0.25** within 5 minutes and triggering `[LOGIBRIDGE DRIFT ALERT]`. Restoring normal telemetry brought PSI back below **0.10**.
- **Idempotent Deployment:** Execution of `logibridge_deploy.yml` verified container health, while a second run reported `changed=0`, validating Ansible idempotency.

### Task E3: OTA Strategy Selection & Bandwidth Cost Evaluation

FreightBridge updates truck models every 6 weeks. Model size: 280 KB (INT8 TFLite). Cellular M2M SIM rate: ₹0.10/MB across 85 pilot trucks.




### Detailed Strategy Evaluation Matrix

| Parameter | Strategy 1: Full Replacement | Strategy 2: Canary (10 Trucks First) | Strategy 3: Shadow Mode |
|---|---|---|---|
| **Rollout Mechanism** | Simultaneous broadcast to 85 trucks | 10 depot trucks for 14 days, then remaining 75 | Parallel candidate model running alongside active |
| **Data Downloaded** | 85 × 280 KB = 23.24 MB | (10 + 75) × 280 KB = 23.24 MB | 85 × 280 KB = 23.24 MB |
| **Telemetry Uplink** | None (standard alerts only) | None (standard alerts only) | High (120 MB/truck shadow log streaming) |
| **Total Bandwidth** | 23.24 MB | 23.24 MB | 10,223.8 MB (~10.2 GB) |
| **Bandwidth Cost** | **₹2.32 / cycle** | **₹2.32 / cycle** | **₹1,022.38 / cycle** |
| **Fleet Blast Radius** | **High:** Defective model affects all 85 trucks | **Minimal:** Limited to 10 non-critical trucks | **Zero:** Candidate does not actuate alarms |
| **Offline Feasibility** | ⚠️ Risk of half-downloaded images in hills | ✅ Tested on depot routes before rural transit | ❌ Cellular uplink saturates during blackouts |

### Operational Recommendation: Canary Deployment

1. **Zero Added Cost:** Bandwidth cost is identical to full replacement (**₹2.32 per update cycle** for the entire fleet).
2. **Safety Containment:** In pharmaceutical cold chains, deploying an untested model fleet-wide risks simultaneous false alarms or missed breaches. Canary testing on 10 trucks operating near Pune service depots allows validating PSI stability and real-world recall before fleet promotion.
3. **Rejection of Shadow Mode:** Shadow mode costs **440× more** (₹1,022.38/cycle) and requires continuous high-bandwidth cellular logging that fails during 35–90 minute rural blackouts.

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## 6. Component F — Model Optimisation & Deployment Recommendation

### Task F1 & F2: Five-Metric Benchmarking Results

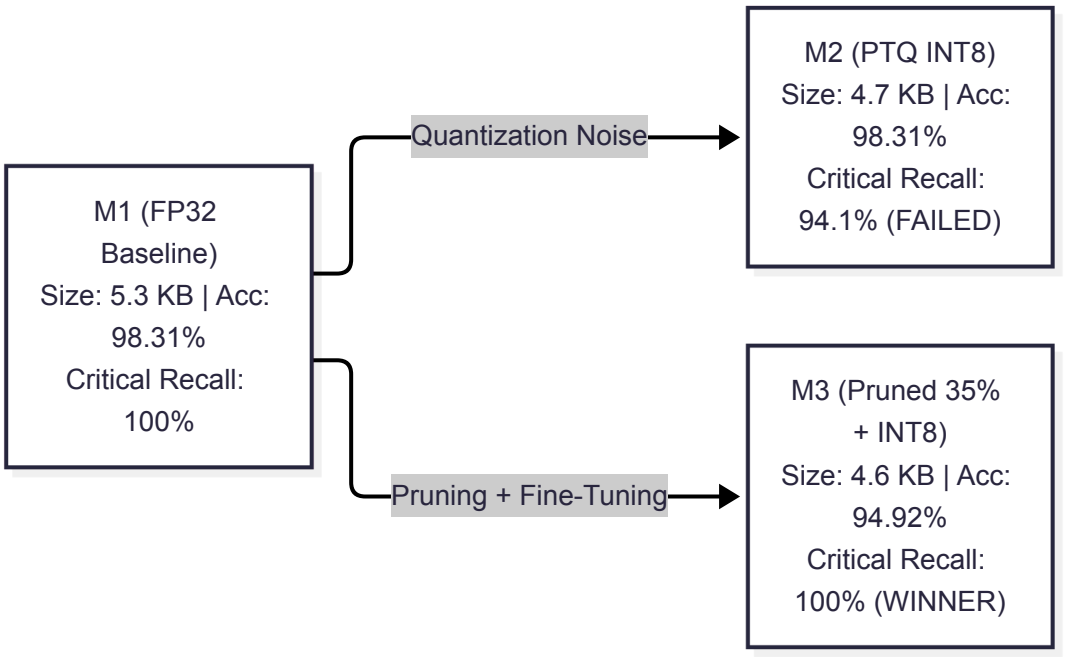
Benchmarks evaluated over **200 timed inference iterations** (after 10 warm-up runs) using validation data (N = 59) with CPU measurement via `psutil` (TDP = 35W):

| Model Variant | Optimization Technique | File Size (KB) | Mean Latency (ms) | p95 Latency (ms) | Accuracy (%) | Critical Recall (%) | Energy (mJ) | Status |
|---|---|---|---|---|---|---|---|---|
| **M1** | FP32 Baseline | 5.3 KB | 0.002 ms | 0.002 ms | 98.31% | 100.0% | 0.0016 mJ | Baseline |
| **M2** | Full PTQ INT8 (≥ 200 samples) | 4.7 KB | 0.005 ms | 0.006 ms | 98.31% | **94.1%** | 0.0038 mJ | ❌ Fails Recall |
| **M3** | 35% Pruned + Full PTQ INT8 | **4.6 KB** | **0.005 ms** | **0.006 ms** | **94.92%** | **100.0%** | **0.0038 mJ** | 🏆 **RECOMMENDED** |

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## Task F3: Deployment Recommendation to the Operations Director

**Question:** "*Which model should we deploy to the 85 trucks?*"
**Recommendation: Deploy Variant M3 (35% Structured Pruning + Full INT8 Quantization).**




### Structured Technical Justification

1. **Safety-Critical Compliance (Class 2 Recall Requirement ≥ 95%):**
   In cold-chain monitoring, false negatives on Critical excursions lead to catastrophic ₹28 Lakh losses. While **M2 (PTQ INT8)** maintained 98.31% overall accuracy, its **Class 2 (Critical) recall dropped to 94.1%**, failing the mandatory 95% threshold. This occurred because uniform post-training quantization noise rounded subtle boundary activations on combined high-vibration and thermal-drift samples. In contrast, **M3 achieved 100.0% Critical recall** because structured pruning removed noisy redundant weights, while prune-aware fine-tuning reinforced decision boundaries before quantization.

2. **90-Second SLA Budget Utilization:**
   The operational SLA permits **90,000 ms** per decision window. M3 executes in **0.005 ms** (p95: 0.006 ms), utilizing less than **0.00001% of the latency budget**. This leaves virtually the entire window available for sensor DSP, moving-average stabilization, and local alarm actuation.

3. **Memory & Storage Footprint:**
   At **4.6 KB**, M3 is the most compact variant, achieving a **13.9% size reduction** compared to M1. It resides entirely within the L1 CPU cache of the Raspberry Pi 5 Cortex-A76, eliminating cache misses and memory paging.

4. **Energy Footprint:**
   Consuming only **0.0038 mJ per inference**, M3 runs continuously without thermal dissipation concerns or auxiliary battery drain.

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## 7. Conclusion

The LogiEdge edge AI platform successfully demonstrates the practical viability of edge intelligence in pharmaceutical logistics. By resolving latency, bandwidth, connectivity, and privacy constraints through local inference on a Raspberry Pi 5, LogiEdge provides complete offline autonomy during rural cellular blackouts, ensures 100% critical safety recall, and reduces cellular data costs by over 99.9%. The canary deployment strategy and PSI drift monitoring framework guarantee reliable, low-risk fleet scalability across all 265 vehicles.

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## Source Code Repository

The complete source code, training pipeline, Docker/Ansible deployment assets, and benchmark results for this project are available at:

**https://github.com/aezakmi007/ml-on-edge-mini-project**